SAYLANI HACKATHON CHALLENGE SPECIFICATION

HCM Lite — Product Requirement Document

Target Date: 24 October 2026

Venue: Zaitoon Ashraf IT Park

Scope: Web Application + AI Assistant + Optional Mobile

9 Modules	25 Features	3 Roles	100% Zero-Data
SYSTEM ARCHITECTURE	CORE CAPABILITIES	ADMIN, HR, EMPLOYEE	FRESH INSTALLATION

1. EXECUTIVE SUMMARY & CORE OBJECTIVES

HCM Lite is an AI-assisted web-based Human Capital Management system designed for small-to-medium enterprise operations. The product streamlines human resources operations across three primary roles: System Admin, HR Managers, and Employees. The platform features an embedded, role-aware AI Assistant operating directly on real system data using structured function calling. The application starts in a completely empty state: company settings, employee rosters, attendance records, and payroll calculations are built entirely from user actions taking place inside the software.

2. PERSONA SPECIFICATIONS & ROLE SCOPE

User Role	Primary Objective	Core System Responsibilities
Admin	Company configuration, governance, and final approval oversight.	First-run installation, office location geofencing, work shifts, leave quotas, HR user provisioning, payroll finalizing (locking), audit log viewing.
HR Manager	Daily workforce operations, employee management, and payroll drafting.	Employee onboarding (CSV bulk import), salary structures, attendance board monitoring, leave request approvals, monthly payroll draft creation.
Employee	Self-service attendance tracking, leave requests, and payslip access.	Web/camera check-in/out, correction requests, AI smart leave generation, personal dashboard monitoring, downloading monthly payslip PDFs.

3. APPLICATION SYSTEM ARCHITECTURE

System Data Flow & Module Interaction

The system relies on a central REST API serving a Single Page Application (SPA) client. Admin configures organization rules (M2), which govern Attendance & Leave checks (M4, M5). Net attendance and unpaid leave calculate monthly Payroll (M6). Dashboards (M7) aggregate these records in real time. The AI Assistant (M8) routes natural language queries directly to the underlying REST service layer, ensuring AI responses perfectly match database facts.

4. DETAILED MODULE SPECIFICATIONS (M1 – M6)

The functional domain of HCM Lite is partitioned into distinct sub-systems. Each module enforces strict server-side permissions and validates input data integrity prior to database writes.

Module	Core Functionality	Business Rules & Validation Logic
M1: Auth	Authentication & RBAC	JWT login with bcrypt hashed passwords. Server-side role validation on every API route. Employees restricted strictly to own `employee_id` data scope. First-run bootstrap creates initial Admin.
M2: Org Setup	Company Rules & Setup	Office start time (09:00), grace period (15 min), half-day threshold (< 5 hrs), weekend definition, office location (latitude, longitude, radius). Applied dynamically to new records.
M3: Employees	Directory & Onboarding	HR onboarding generates employee login credentials with temporary passwords. Server-side pagination, search, filter. Optional CSV import with dry-run report flagging exact row/field errors.
M4: Attendance	Time Tracking	Single record per employee per day (duplicate returns `409 Conflict`). Server-stamped time. Automatic status tag: Present (≤ 09:15), Late (> 09:15), Half-day (< 5 hrs), Absent (no record).
M5: Leave	Leave Processing	Default annual quotas: Annual (14), Sick (8), Casual (10), Unpaid (unlimited). Excludes weekends and public holidays automatically. Date overlaps or quota overflow returns `422 Unprocessable`.
M6: Payroll	Salary & Payslips	Gross = 1.5 × Basic. Deductions calculated on daily rate for absent and unpaid days. Progressive annual tax slab evaluation. HR creates draft; Admin finalizes to permanently lock month.

5. TAX SLABS & SALARY LOGIC

> **Illustrative Annualized Tax Structure**
>
> - **0 to 600,000 PKR:** 0% tax rate.
> - **> 600,000 to 1,200,000 PKR:** 5% of amount exceeding 600,000 PKR.
> - **> 1,200,000 to 2,400,000 PKR:** 30,000 PKR + 15% of amount exceeding 1,200,000 PKR.
> - **> 2,400,000 to 3,600,000 PKR:** 210,000 PKR + 25% of amount exceeding 2,400,000 PKR.
> - **> 3,600,000 PKR:** 510,000 PKR + 35% of amount exceeding 3,600,000 PKR.

6. AI ASSISTANT ARCHITECTURE (MODULE 8)

Security, Guardrails & Execution Flow

The AI Assistant interfaces via hosted LLM APIs with tool calling capabilities. The browser client never holds the LLM key. All calls route through POST `/ai/chat` on the backend server.

- **Tool Execution:** Models invoke backend routines (`get_headcount`, `get_attendance_summary`, `get_late_comers`, `get_leave_summary`, `get_my_leave_balance`, `get_payroll_summary`, `search_employees`).
- **RBAC Scoping:** AI tool execution enforces identical permissions as user API endpoints (e.g., salary queries for peers are rejected).
- **Smart Leave:** Converts natural language ("Need next Mon and Tue off for wedding") into pre-filled leave JSON for confirmation.
- **Guardrails:** 15-second timeout, maximum 20 requests/user/hour, 1 retry on JSON parse failure, mandatory "AI-generated" UI tag.

7. CORE ANALYTICS & KEY PERFORMANCE INDICATORS

Metric Name	Mathematical Formula	Dashboard Visibility
Attendance Rate	`(Present + Late + Missing_Checkout + 0.5 × Half_Day) / (Working Days - On Leave)`	HR Dashboard, Admin Dashboard
Late Percentage	`Late Days / (Present Days + Late Days) × 100`	HR Dashboard (by Department)
Absenteeism Rate	`Absent Days / Expected Working Days × 100`	HR Dashboard (by Department)
Attrition Rate	`Exits in Period / Average Headcount × 100`	Admin Dashboard (YTD)

8. OPTIONAL CAMERA ATTENDANCE & GEOFENCING ARCHITECTURE

Step	Client Action (Web / Mobile)	Server Verification Rule
1. Camera Capture	Opens front camera feed, runs on-device face detector.	Rejects if 0 faces or > 1 face detected in frame.
2. Location Read	Reads device GPS coordinates + reported accuracy.	Rejects if reported GPS accuracy > 100 meters.
3. Geofence Check	Transmits coordinates to backend check-in endpoint.	Calculates distance using Haversine formula; rejects if outside radius.

9. MASTER REQUIREMENTS & RBAC MATRIX (25 FEATURES)

#	Functionality / Requirement	Admin	HR	Employee
01	First-run setup: company profile and Admin creation	Manage	—	—
02	Login, logout, change password; role-based menus	Own	Own	Own
03	Company settings: office hours, grace, weekend, geofence	Manage	View	—
04	Departments, designations, leave quotas, public holidays	Manage	View	—
05	Create HR logins; deactivate or reset any user credential	Manage	—	—
06	Register employee (creates login) & edit details	View	Manage	Own Phone/Addr
07	Employee directory with search/filters; 360 profile	All	All	Own
08	CSV bulk import with dry-run error reporting; CSV export	—	Manage	—
09	Check in and check out (Web / Mobile)	—	Own	Own
10	Attendance calendar & daily attendance board	All	All	Own
11	Attendance correction: submit request and approve	View	Approve	Request
12	Apply / cancel leave; view balance and history	—	Own	Own
13	Approve or reject leave request with comment	View	Approve	—
14	Salary structure configuration per employee	View	Manage	—
15	Monthly payroll: generate and review draft run	View	Manage	—
16	Finalize payroll (permanently locks the month)	Manage	—	—
17	Payslip generation with PDF download	All	All	Own
18	Dashboards, KPIs, trends, and analytical insight lists	Company	HR	Personal
19	AI Assistant chat with tool calling on real data	All Data	HR Data	Own Data
20	AI dashboard analytical summary widget	Yes	Yes	—
21	Smart leave: natural text parsing to pre-filled form	Yes	Own	Own
22	Policy Q&A on attendance, leave, and payroll rules	Yes	Yes	Yes
23	Announcements publishing	Post	Post	Read
24	In-app notifications (leave decisions, payslips)	Own	Own	Own
25	System-wide audit log of administrative actions	View	—	—